



Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Advanced Machine Learning

Roll No.:

Course Code: CSPC-611(vi)

Examination: 2nd Periodical Examination

Scheme: NEP

Branch: CSE (AI & ML)

Semester: VI

Question Paper for 2nd Periodical Examination (May 2026) (Session Jan-June 2026)

Time: One and a half hours

Max Marks:30

Note: Attempt all questions.

Date: 08.05.2026(10-11:30 AM)

CO	COURSE OUTCOMES
CO1	Design and implement machine learning solutions to classification, regression and clustering problems.
CO2	Evaluate and interpret the results of the different ML techniques.
CO3	Design and implement various machine learning algorithms in a range of Real-world applications.

Q No.	Questions	Marks	Target CO	Target KL
1	Attempt all:	10	1,2,3	
1.a	Differentiate between conditional probability and joint probability	2	CO3	L3
1.b	Define reinforcement learning.	2	CO4	L2
1.c	Differentiate analytical and numerical gradients.	2	CO3	L3
1.d	Define Bayesian nets with an example.	4	CO3	L3
2.	Explain gradient checking with an example.	10	CO2	L3
3	Explain any two weight initialization methods.	5	CO2	L3
4.	Explain the decision tree and its functionality with an example.	5	CO4	L3

KL: Knowledge Level: L1 - Remember L2 - Understand L3 - Apply L4 - Analyze L5 - Evaluate L6 - Create
Course Advisor: Shivani (Assistant Professor, CSE)